

Omer Kilic

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Aerospace Engineering · Computational Aerodynamics · Scientific Computing · Manufacturing & Testing · Propulsion

EDUCATION

Rutgers, The State University of New Jersey · New Brunswick, NJ · B.S. Aerospace Engineering Expected May 2028
GPA 4.00/4.00 · Aresty Research Scholar · Richard Peters & Cecilia Santiago Peters Endowed Scholar

Bergen Community College · Paramus, NJ · A.S. Engineering Science 2025
GPA 3.90/4.00 · Phi Theta Kappa · USDA 3SP Scholar (\$3,000)

Bilkent University · Ankara, Turkiye 2018 – 2020
Economics coursework · Full scholarship covering tuition and stipend · National University Entrance rank: **1,085 of 2.5M** candidates

TECHNICAL SKILLS

Aerodynamics & Numerics Fast Multipole Method, discrete-vortex methods, unsteady wake modeling, multipole expansions, 6-DOF store-separation trajectory analysis, flight dynamics and stability derivatives, CFD meshing and post-processing

Scientific Computing Julia, Python (NumPy, Pandas, Matplotlib, Jupyter), MATLAB, C++, Git, LaTeX/Typst

Optimization & Verification automatic differentiation, gradient-based design optimization, Monte Carlo / GCMC, surrogate and reduced-order modeling, numerical verification, error analysis and convergence study

Design & Manufacturing SolidWorks, composites manufacturing, rotor/propeller balancing, GD&T, NX/Fibersim workflow exposure, NCR/MRB

Test & Instrumentation wind-tunnel test exposure, sensor monitoring and diagnostics, CAN Bus, NEC code compliance

RESEARCH & AEROSPACE EXPERIENCE

Aresty Research Assistant · Flapping-Wing Aerodynamics Lab, Rutgers Mechanical and Aerospace Engineering Department. 2026 – Present
PI: Prof. Mitsunori Denda · Bio-Inspired Micro Aerial Vehicles · insectflight.rutgers.edu

- Built a Fast Multipole Method in Julia for a discrete-vortex flapping-wing solver and integrated it into the group's MATLAB 3-D wake code as a replacement velocity kernel; with vectorized exact evaluation, cut a 15-step simulation from **1,783 s to 50 s (35×)**.
- Derived multipole expansions for vortex rings and polygonal panels and built the full FMM pipeline with a priori far-field error bounds, verified against direct summation across **39 automated test groups**.
- Made the solver differentiable through automatic differentiation for gradient-based flapping-stroke design, and characterized the horizon over which gradients remain reliable.
- Authored a 10-part FMM tutorial series and group presentations; built and maintain the lab's hub, flapping-wings.vercel.app.

Candidate Engineer · SKY Global Turk Internship Program · Turkish Aerospace Industries (TUSAŞ) · Ankara, Turkiye Aug – Sep 2026
Turkiye Internship Program (YTB) · selected via technical interview · UAV Manufacturing & KAAN Aerodynamics

- Ran [KAAN](#) store-separation aerodynamic studies end to end and reported the results: geometry preparation, meshing, solver execution, 6-DOF trajectory analysis, post-processing; observed scaled-store wind-tunnel tests across multiple configurations.
- Delivered a first-principles flight-dynamics analysis of the KAAN fighter: derived the 6-DOF Newton–Euler equations of motion, stability derivatives and spin behavior and documented how the aerodynamic database and surrogate models are built.
- Delivered a technical report on T129 ATAK rotor-blade manufacturing and balancing to evaluate production of AKSUNGUR UAV propeller blades (master tooling, tolerances, CG control), including gram-scale tip trim-mass corrections and their moment-arm effect on balance and vibration.
- Documented the design-to-production data chain (composite manufacturing, Teamcenter routing, NX/Fibersim ply data, NCR/MRB) across the AKSUNGUR, ANKA-3, Super Şimşek and AYAZ programs, analyzing process consistency

Research Intern · BCC-NJIT Summer Research Program · Laboratory for Materials Interfaces (Prof. Gennady Gor), Otto H. York Department of Chemical and Materials Engineering · New Jersey Institute of Technology · Newark, NJ Summer 2025
– Ran Grand Canonical Monte Carlo simulations of helium confined in zirconium nanovoids at 573 K and 973 K (Mezcal), establishing thermodynamic baselines against irradiation literature; built Python/Jupyter workflows automating runs, equilibrium assessment and density/pressure-vs- μ analysis.

PROJECTS & ADDITIONAL EXPERIENCE

Founder & Electrical Technician · Hi Tech Electrical Contractors Inc. · Palisades Park, NJ 2020 – Present
Electrician since 2020; incorporated the company in 2023 and operate it while studying full time; perform panel upgrades, branch-circuit wiring, NEC-compliant installations, and fault diagnostics. Home Depot Pro Referral partner with **22 verified five-star reviews** (5.0/5.0). [\[Link\]](#).

Project Member · Electric Vehicle Conversion · Bergen Community College STEM Research Center · Paramus, NJ 2024 – 2025
Converted a 1998 Chevrolet to an electric drivetrain; developed a CAN Bus sensor monitoring and fault-diagnostic system.

Member · Rutgers Propulsion Lab (AIAA) · New Brunswick, NJ 2026 – Present
Contributed CAD models for structural components for student rocket development; investigated paraffin-based hybrid rocket fuels.

SERVICE & VOLUNTEERING

Environmental Service Volunteer · USDA 3SP Scholarship Program Bergen Community College
Completed a semester of environmental service, including campus tree planting and invasive-species removal; received a \$3,000 USDA 3SP scholarship.

INTERESTS

Computational Aerospace | Biomimetic Flight Mechanics | Swimming | Fitness | Aviation History | Road Trips